Kernel Tricks

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Outline

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Kernel

A symmetric, real-valued function $K : T \times T \to \mathbb{R}$ is said to be **positive definite** if, for all $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$ and $t_1, \ldots, t_n \in T$

$$\sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j K(t_i, t_j) \geq 0$$

is satisfied for any $n \geq 1$. Here, we call the function K symmetric if it satisfies

$$K(t, s) = K(s, t)$$

for any $t, s \in T$. Here, T may be regarded as a compact subset of $\mathbb{R}^p$.

Kernel

The usual inner product $\langle t_i, t_j \rangle = t_i' t_j$ defined in $\mathbb{R}^p$ is a positive definite kernel. To see this, note that for all $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$ and $t_1, \ldots, t_n \in T$, we have

$$\sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j t_i' t_j = \left(\sum_{i=1}^{n} \alpha_i t_i \right)' \left(\sum_{j=1}^{n} \alpha_j t_j \right) \geq 0$$

Other examples of positive definite kernels include the radial basis function kernels

$$K(t, s) = e^{-\gamma \|t-s\|^2}, \quad \gamma > 0$$

and polynomial kernel

$$K(t, s) = (\alpha \langle t, s \rangle + 1)^d, \quad d \in \mathbb{N}, \alpha > 0$$

Kernel trick

Kernel trick: Formulate an estimation problem only in terms of inner products, i.e.,

$$\langle x_i, x_j \rangle = x_i' x_j$$

for $i, j = 1, \ldots, n$. Replace the inner product $x_i' x_j$ with a positive definite kernel $K(x_i, x_j)$ for all $i, j = 1, \ldots, n$.

Kernel tricks for linear regression models

Original formulation of OLS

Suppose $\mathbf{X}$ is $n \times p$ dimensional data matrix. Recall that OLS estimator is defined as

$$\hat{\beta}^{ols} = \arg\min_{\beta} (\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta)$$

1. Suppose $p < n$. Then, $(\mathbf{X}'\mathbf{X})^{-1}$ exists, and we have a unique solution given by

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

2. Suppose $p > n$. Then, $(\mathbf{X}'\mathbf{X})^{-1}$ does not exist, and we do not have a unique solution. Any $\hat{\beta}$ such that the squared error becomes zero is a solution.

Original formulation of OLS

2. (Conti.) The solution is characterized by the equation

$$\mathbf{Y} = \mathbf{X}\hat{\beta}$$

Let $\mathbf{X} = UDV'$ be the singular value decomposition of $\mathbf{X}$. Then, $\hat{\beta}$ given by

$$\begin{aligned}\hat{\beta} &= \mathbf{X}^{+}\mathbf{Y} \\ &= VD^{-1}U'\mathbf{Y}\end{aligned}$$

is a solution. Here, $\mathbf{X}^{+} = VD^{-1}U'$ is the Moore-Penrose inverse of $\mathbf{X}$ and $\hat{\beta} = VD^{-1}U'Y$ gives the minimum (Euclidean) norm solution to $\mathbf{Y} = \mathbf{X}\hat{\beta}$.

Original formulation of OLS

2. (Conti.) The full set of solutions can be written as

$$\hat{\beta} = VD^{-1}U'\mathbf{Y} + v$$

where v is any p dimensional vector in the null space of $\mathbf{X}$,i.e.,

$$\{v \in \mathbb{R}^p : \mathbf{X}v = 0\}$$

The squared norm of $\hat{\beta}$ is given by

$$\begin{aligned}\|\hat{\beta}\|^2 &= \mathbf{Y}UD^{-1}V'VD^{-1}U'\mathbf{Y} + 2v'VD^{-1}U'\mathbf{Y} + v'v \\ &= \mathbf{Y}UD^{-1}V'VD^{-1}U'\mathbf{Y} + v'v\end{aligned}$$

where the second equality follows because the null space of $\mathbf{X}$ is equivalent to the null space of V. Thus, the minimum norm estimator $\hat{\beta}$ is given by taking $v = 0$.

Inner product fomulation of OLS

Let us formulate the OLS as

$$\hat{\alpha}^{ols} = \text{arg} \min_{\alpha \in \mathbb{R}^n} (\mathbf{Y} - \mathbf{X}\mathbf{X}'\alpha)'(\mathbf{Y} - \mathbf{X}\mathbf{X}'\alpha)$$

Here, $\mathbf{X}\mathbf{X}'$ is what's called the Gram matrix and is given by

$$\mathbf{X}\mathbf{X}' = \begin{pmatrix} x_1' \\ \vdots \\ x_n' \end{pmatrix} \begin{pmatrix} x_1 & \dots & x_n \end{pmatrix}$$

$\mathbf{X}\mathbf{X}'$ is $n \times n$ dimensional matrix with $x_i' x_j$ as its (i, j)-th element. The inner product formulation is only done so that we may use the kernel trick.

Inner product formulation of OLS

$$\hat{\alpha}^{ols} = \text{arg} \min_{\alpha \in \mathbb{R}^n} (\mathbf{Y} - \mathbf{X}\mathbf{X}'\alpha)'(\mathbf{Y} - \mathbf{X}\mathbf{X}'\alpha)$$

After α has been estimated, we'd predict a new sample $\hat{y}_{n+1}$ using the formula

$$\begin{aligned}
\hat{y}_{n+1} &= x'_{n+1}\mathbf{X}'\hat{\alpha} \\
&= x'_{n+1}\left(\sum_{i=1}^{n} x_i\hat{\alpha}_i\right) \\
&= \sum_{i=1}^{n} \hat{\alpha}_i x'_{n+1} x_i
\end{aligned}$$

Thus, we may regard

$$\hat{f}(x) = \sum_{i=1}^{n} \hat{\alpha}_i x' x_i$$

as an estimator for the conditional mean function $f(x) = \mathbb{E}[y|X = x]$.

Inner product formulation of OLS

1. Suppose $n > p$. Then, the solution $\hat{\alpha}$ is not unique and any $\hat{\alpha}$ that satisfies

$$\mathbf{X}'\hat{\alpha} = \hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

is a solution. The minimum norm solution to the above equation is

$$\begin{aligned}
\hat{\alpha} &= (\mathbf{X}')^{+}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} \\
&= UD^{-1}V'(VDU'UDV')^{-1}VDU'\mathbf{Y} \\
&= UD^{-1}V'(VD^{-2}V')VDU'\mathbf{Y} \\
&= UD^{-2}U'\mathbf{Y}
\end{aligned}$$

and the full set of the solution is characterized by

$$\hat{\alpha} = UD^{-2}U'\mathbf{Y} + v$$

for some vector v in the null space of $\mathbf{X}'$. Here, $(\mathbf{X}\mathbf{X}')^{+} = UD^{-2}U'$ is the Moore-Penrose inverse of $\mathbf{X}\mathbf{X}'$.

Inner product formulation of OLS

2. Suppose $n < p$. Then, the solution $\hat{\alpha}$ is unique and is given by

$$\hat{\alpha} = (\mathbf{X}\mathbf{X}')^{-2}\mathbf{X}\mathbf{X}'\mathbf{Y} = (\mathbf{X}\mathbf{X}')^{-1}\mathbf{Y}$$

In this case, $\hat{\beta}$ is not unique. However, we still have

$$\mathbf{X}\hat{\beta} = \mathbf{X}\mathbf{X}'\alpha = \mathbf{Y}$$

Thus, we have a perfect in-sample fit in both the original and inner product formulation of OLS. If we predict $\hat{y}_{n+1}$ with $\hat{y}_{n+1} = x'_{n+1}\mathbf{X}'\hat{\alpha} = x'_{n+1}\mathbf{X}'(\mathbf{X}\mathbf{X}')^{-1}\mathbf{Y}$. This is equivalent to the prediction obtained in the original formulation if we use the minimum norm solution $\hat{\beta} = VD^{-1}U'\mathbf{Y}$ because

$$\mathbf{X}'\hat{\alpha} = \mathbf{X}'(\mathbf{X}\mathbf{X}')^{-1}\mathbf{Y} = VD^{-1}U'\mathbf{Y} = \hat{\beta}$$

Kernel Trick

Kernel trick in OLS is done by replacing $x_i' x_j$ with $K(x_i, x_j)$ in the inner product formulation of OLS.

$$\hat{\alpha}^{kernel,ols} = \text{arg} \min_{\alpha \in \mathbb{R}^n} (\mathbf{Y} - \mathbf{K}\alpha)'(\mathbf{Y} - \mathbf{K}\alpha)$$

where $\alpha = (\alpha_1, \ldots, \alpha_n)'$ and $\mathbf{K}$ is an $n \times n$ dimensional matrix with $K(x_i, x_j)$ as its (i, j)-th element. The solution is always unique (if we are using a strictly positive definite kernel) and is given by

$$\hat{\alpha}^{kernel,ols} = \mathbf{K}^{-1}\mathbf{Y}$$

We always get a perfect fit. If we are given a new sample x_{n+1}, the predicted value will be given by

$$\hat{y}_{n+1} = \sum_{i=1}^{n} \hat{\alpha}_i K(x_{n+1}, x_i)$$

However, we usually wouldn't want to use this model for prediction because $\mathbf{K}^{-1}$ is not well-behaved.

Original formulation of ridge regression

The ridge regression is given by

$$\hat{\beta} = \text{arg}\min_{\beta \in \mathbb{R}^p} (\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta) + \lambda \beta' \beta$$

If $\lambda > 0$, the solution always exists uniquely as

$$\hat{\beta} = (\mathbf{X}'\mathbf{X} + \lambda I_p)^{-1}\mathbf{X}'\mathbf{Y}$$

Inner product formulation of ridge regression

Consider the inner product formulation of ridge regression

$$\hat{\alpha} = \text{arg}\min_{\alpha \in \mathbb{R}^n} (\mathbf{Y} - \mathbf{X}\mathbf{X}'\alpha)'(\mathbf{Y} - \mathbf{X}\mathbf{X}'\alpha) + \lambda\alpha'\mathbf{X}\mathbf{X}'\alpha$$

As before, we predict $\hat{y}_{n+1}$, using

$$\begin{aligned}\hat{y}_{n+1} &= x'_{n+1}\hat{\beta} \\ &= x'_{n+1}\mathbf{X}'\hat{\alpha} \\ &= \sum_{i=1}^{n} \hat{\alpha}_i x'_{n+1} x_i\end{aligned}$$

The FOC is given by

$$-\mathbf{X}\mathbf{X}'(\mathbf{Y} - \mathbf{X}\mathbf{X}'\alpha) + \mathbf{X}\mathbf{X}'\alpha = 0$$

so that the solution is characterized by

$$\mathbf{X}\mathbf{X}'\mathbf{Y} = \mathbf{X}\mathbf{X}'(\mathbf{X}\mathbf{X}' + \lambda I_n)\alpha$$

Inner product formulation of ridge regression

The solution is characterized by

$$\mathbf{X}\mathbf{X}'\mathbf{Y} = \mathbf{X}\mathbf{X}'(\mathbf{X}\mathbf{X}' + \lambda I_n)\alpha$$

1. Suppose $n > p$. Then, $(\mathbf{X}\mathbf{X}')^{-1}$ does not exist. The solution is not unique and the solution is characterized by

   $$\hat{\alpha} = (\mathbf{X}\mathbf{X}' + \lambda I_n)^{-1}\mathbf{Y} + v$$

   for some v in the null space of $\mathbf{X}'$.
2. Suppose $n < p$. Then, $(\mathbf{X}\mathbf{X}')^{-1}$ exists and the unique solution is given by

   $$\hat{\alpha} = (\mathbf{X}\mathbf{X}' + \lambda I_n)^{-1}\mathbf{Y}$$

Inner product formulation of ridge regression

Note that we have

$$(\mathbf{X}'\mathbf{X} + \lambda I_p)\mathbf{X}' = \mathbf{X}'(\mathbf{X}\mathbf{X}' + \lambda I_n)$$

pre- and post- multiplying both sides by $(\mathbf{X}\mathbf{X}' + \lambda I_n)^{-1}$, we obtain

$$(\mathbf{X}'\mathbf{X} + \lambda I_p)^{-1}\mathbf{X}' = \mathbf{X}'(\mathbf{X}\mathbf{X}' + \lambda I_n)^{-1}$$

Thus, regardless of the size of n, p, we always have

$$\begin{aligned}\mathbf{X}'\hat{\alpha} &= \mathbf{X}'(\mathbf{X}\mathbf{X}' + \lambda I_n)^{-1}\mathbf{Y} \\ &= (\mathbf{X}'\mathbf{X} + \lambda I_p)^{-1}\mathbf{X}' \\ &= \hat{\beta}\end{aligned}$$

Kernel Trick

In the inner product formulation of ridge regression, we replace $\mathbf{X}\mathbf{X}'$ with $\mathbf{K}$. Then, we have

$$\hat{\alpha} = \text{arg} \min_{\alpha \in \mathbb{R}^n} (\mathbf{Y} - \mathbf{K}\alpha)'(\mathbf{Y} - \mathbf{K}\alpha) + \lambda \alpha' \mathbf{K} \alpha$$

where $\alpha = (\alpha_1, \ldots, \alpha_n)'$ and $\mathbf{K}$ is an $n \times n$ dimensional matrix with $K(x_i, x_j)$ as its (i, j)-th element. The solution is always unique and is given by

$$\hat{\alpha} = (\mathbf{K} + \lambda I_n)^{-1} \mathbf{Y}$$

If we are given a new sample x_{n+1}, the predicted value will be given by

$$\hat{y}_{n+1} = \sum_{i=1}^{n} \hat{\alpha}_i K(x_{n+1}, x_i)$$

Original formulation of PCA

Let P_ι be $n \times n$ dimensional projection matrix given by

$$P_\iota = \iota_n(\iota_n' \iota_n)^{-1} \iota_n'$$

and $M_\iota = I_n - P_\iota$. Then,

$$\frac{1}{n}\mathbf{X}'(I_n - P_\iota)\mathbf{X}$$

is $p \times p$ dimensional sample variance. PCA sequentially finds the direction $v \in \mathbb{R}^p$ such that the linear combination $v' x_i$ has maximum sample variance.

Original formulation of PCA

The sample variance of $v'x_i$ is given by

$$\frac{1}{n}\sum_{i=1}^{n}(v'x_i - v'\bar{x})^2 = \frac{1}{n}\sum_{i=1}^{n} v'(x_i - \bar{x})(x_i - \bar{x})'v$$
$$= \frac{1}{n} v'\mathbf{X}'(I_n - P_\iota)\mathbf{X}v'$$

Thus, the first problem is written as

$$\max_{v \in \mathbb{R}^p} v'\mathbf{X}'(I_n - P_\iota)\mathbf{X}v \quad \text{s.t. } v'v = 1$$

and solution is given by the first eigenvector v_1 of $\mathbf{X}'(I_n - P_\iota)\mathbf{X}$. We denote $f_{1i} = v_1' x_i$ as the first principal component.

Original formulation of PCA

The next problem is given by

$$\max_{v \in \mathbb{R}^p} v'\mathbf{X}'(I_n - P_\iota)\mathbf{X}v \quad \text{s.t. } v_1'v = 0, v'v = 1$$

and the solution is given by the second eigenvector v_2 of $\mathbf{X}'(I_n - P_\iota)\mathbf{X}$ and we denote $f_{2i} = v_2'x_i$ as the second principal component. We may proceed in this way until all p number of principal components $(f_{1i}, \ldots, f_{pi})'$ are found.

Original formulation of PCA

Let the singular value decomposition of $(I_n - P_\iota)\mathbf{X}$ be given by

$$(I_n - P_\iota)\mathbf{X} = UDV'$$

Then, the $n \times p$ dimensional matrix collecting the demeaned principal components is given by

$$UD = (I_n - P_\iota)\mathbf{X}V = \begin{pmatrix} x_1' - \bar{x}' \\ \vdots \\ x_n' - \bar{x}' \end{pmatrix} V = \begin{pmatrix} (x_1' - \bar{x}')V \\ \vdots \\ (x_n' - \bar{x}')V \end{pmatrix} = \begin{pmatrix} x_1'V - \bar{x}'V \\ \vdots \\ x_n'V - \bar{x}'V \end{pmatrix} = \begin{pmatrix} f_1' - \bar{f}' \\ \vdots \\ f_n' - \bar{f}' \end{pmatrix}$$

because

$$x_i'V = x_i'(v_1, \ldots, v_p) = (f_{1i}, \ldots, f_{pi}) = f_i'$$

and

$$\bar{x}'V = \left(\frac{1}{n}\sum_{i=1}^{n} x_i'\right) V = \frac{1}{n}\sum_{i=1}^{n} f_i' = \bar{f}'$$

Original formulation of PCA

PCA gives a basis expansion of $x_i \in \mathbb{R}^p$ as

$$x_i = \sum_{j=1}^{p} (v_j' x_i) v_j$$

where the sample variance of $f_{ji} = v_j' x_i$ are decreasing in j and where $f_{ji} = v_j' x_i$ and $f_{li} = v_l' x_i$ are orthogonal for $j \neq l$.

Inner product formulation of PCA

Recall the singular value decomposition $(I_n - P_\iota)\mathbf{X} = UDV'$. We can formulate the estimation of PCA only in terms of $\mathbf{X}\mathbf{X}'$. Notice that from

$$(I_n - P_\iota)\mathbf{X}'\mathbf{X}(I_n - P_\iota) = UD^2U'$$

we may compute

$$\begin{pmatrix} f_1' - \bar{f}' \\ \vdots \\ f_n' - \bar{f}' \end{pmatrix} = UD$$

Kernel trick

Replace $\mathbf{XX}'$ with $\mathbf{K}$. Let the spectral representation be given by

$$(I_n - P_\iota)\mathbf{K}(I_n - P_\iota) = \tilde{U}\tilde{D}^2\tilde{U}'$$

We can compute the kernel principal components as

$$\begin{pmatrix} f_1' - \bar{f}' \\ \vdots \\ f_n' - \bar{f}' \end{pmatrix} = \tilde{U}\tilde{D}$$

Kernel tricks for linear classification models

Linear SVM

The dual Problem of the linear SVM is given by

$$\max_{(\alpha_i)} g(\alpha)\{=\sum_{i=1}^{n}\alpha_i - \frac{1}{2}\sum_{i=1}^{n}\sum_{k=1}^{n}\alpha_i\alpha_k y_i y_k x_i' x_k\} \text{ s.t. } \begin{cases} 0 = \sum_{i=1}^{n}\alpha_i y_i \\ 0 \leq \alpha_i \leq C \quad \forall i = 1,\dots,n \end{cases}$$

with the decision function

$$\begin{aligned} f(x) &= \beta_0 + x'\beta \\ &= \beta_0 + \sum_{i=1}^{n}\alpha_i y_i x' x_i \end{aligned}$$

because we have $\beta = \sum_{i=1}^{n}\alpha_i y_i x_i$ at the optimum.

(Kernel) SVM

The dual Problem of the linear SVM is given by

$$\max_{(\alpha_i)} g(\alpha)\{= \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{k=1}^{n} \alpha_i \alpha_k y_i y_k K(x_i, x_k)\} \text{ s.t. } \begin{cases} 0 = \sum_{i=1}^{n} \alpha_i y_i \\ 0 \leq \alpha_i \leq C \quad \forall i = 1, \ldots, n \end{cases}$$

with the decision function

$$\begin{aligned} f(x) &= \beta_0 + x'\beta \\ &= \beta_0 + \sum_{i=1}^{n} \alpha_i y_i K(x, x_i) \end{aligned}$$

because we have $\beta = \sum_{i=1}^{n} \alpha_i y_i x_i$ at the optimum.

Kernelized Logit Model

Primal Problem:

$$\min_{\beta,\beta_0} \sum_{i=1}^{n} \log(1 + e^{-y_i(x_i'\beta+\beta_0)}) + \lambda\|\beta\|_2^2$$

Dual Problem:

$$\max_{\alpha_i} -C\sum_{i=1}^{n} G\left(\frac{\alpha_i}{C}\right) - \frac{1}{2}\sum_{i=1}^{n}\sum_{k=1}^{n} \alpha_i\alpha_k y_i y_k x_i' x_k \quad \text{s.t.} \quad \sum_{i=1}^{n} \alpha_i y_i = 0$$

where $G(\delta) = \delta\log(\delta) + (1-\delta)\log(1-\delta)$, and the decision function is given by

$$\begin{aligned} f(x) &= x'\beta + \beta_0 \\ &= \sum_{i=1}^{n} \alpha_i y_i x_i' x + \beta_0 \end{aligned}$$

where we've used $\beta = \sum_{i=1}^{n} \alpha_i y_i x_i$, which is satisfied at the optimum.

Basis Expansion

Basis Expansion

How do we go from linear models to nonlinear models? One popular way is to enlarge the original space of X through some nonlinear transformations $h_m(X) : \mathbb{R}^p \to \mathbb{R}$, $m = 1, \ldots, M$. Then, we model $f(X)$ using a linear model in $h(X)$.

$$f(X) = \sum_{m=1}^{M} \beta_m h_m(X)$$

Examples:

1. $h_m(X) = X_j^2$ or $h_m(X) = X_j X_i$: Quadratic term or terms capturing interactions.
2. $h_m(X) = 1_{\{X_j \geq c_m\}}$: Indicator functions

Here, we assume that $h_m(X), m = 1, \ldots, M$ is known and fixed. While $f(X)$ is nonlinear in X, note that it is still linear in β_m.

Basis Expansion

Using the transformed variables $h(X) = (h_1(X), \dots, h_M(X))'$, we may perform OLS, ridge regression, PCA, SVM, or logistic regression.
OLS:

$$\hat{\beta}^{ols} = \operatorname{arg}\min_{\beta} \sum_{i=1}^{n} (Y_i - \beta' h(X_i))^2$$

Ridge:

$$\hat{\beta}^{ridge} = \operatorname{arg}\min_{\beta} \sum_{i=1}^{n} (Y_i - \beta' h(X_i))^2 + \lambda \beta' \beta$$

Basis Expansion

PCA:

$$\hat{v} = \arg\max_{v \in \mathbb{R}^M} v' \left(\sum_{i=1}^{n} h(X_i) h(X_i)' \right) v$$

SVM:

$$\min_{\beta, \beta_0, (\xi_i)} \frac{1}{2} \|\beta\|^2 + C \sum_{i=1}^{n} \xi_i \quad \text{s.t.} \quad \begin{cases} \xi_i \geq 1 - y_i(\beta' h(x_i) + \beta_0) \\ \xi_i \geq 0, \quad i = 1, \ldots, n \end{cases}$$

(Penalized) Logit:

$$\min_{\beta, \beta_0} \sum_{i=1}^{n} \log(1 + e^{-y_i(\beta' h(x_i) + \beta_0)}) + \lambda \beta' \beta$$

Kernel Trick

Recall that all of the estimation problems given above can be written only in terms of inner products $\langle h(X_i), h(X_j)\rangle$. Instead of specifying the basis $h_m(X), m = 1, \dots, M$ directly, let's only specify how the inner product of $h(X)$ behaves. We specify $h(X)$ as some function $h : \mathbb{R}^p \to \mathbb{R}^M$ that satisfies

$$\langle h(X_i), h(X_j)\rangle = K(X_i, X_j)$$